A large-scale investigation of pronoun interpretation biases in LLMs

Anonymous
Anonymous University
Anonymous Email Address

Humans have strong intuitions about pronoun interpretation even in the absence of much information [1–4]:

1. Alfred frightened Bart, because he... [*he* = Alfred]
2. Alfred liked Bart, because he... [*he* = Bart]
3. Because Alfred frightened Bart, he... [*he* = Bart]
4. Because Alfred liked Bart, he... [*he* = Alfred]

Are these pronoun biases a phenomenon in their own right (e.g., perhaps they exist to speed real-time comprehension), or do they fall out of general pronoun processing mechanisms? A partial answer could come from Large Language Models (LLMs), which perform quite well on difficult pronoun interpretation tasks [5, 6] but are generally not trained for pronoun biases.

A number of studies have investigated pronoun biases in LLMs, with variable results [7–11]. Some focused on the 305 verbs reported in [12], though others tested only a few dozen verbs. Because there is a long history in pronoun bias research of reporting results that did not generalize to additional stimuli [13], we return to this question with a much larger data set. We also test newer LLMs; most published studies use GPT-2 [which performs poorly on pronouns; 14] or similar.

We created a large dataset by combining two public datasets of human judgments [2, 13], which in total contain human pronoun bias results for 1,484 **implicit causality** sentences [similar to (1-2)] involving 1,212 distinct verbs, and 501 **implicit consequentiality** sentences [similar to (3-4)] involving 501 distinct verbs. We determined the noise ceiling (the best correlation between model and human data we could achieve, given statistical noise in the human data) by comparing two synthetic human datasets obtained by resampling the data with replacement. The noise ceiling was $r = .91$ for implicit causality and $r = .66$ for implicit consequentiality (far fewer participants had provided judgments for the implicit consequentiality dataset).

We tested two transformer-based LLMs. **T5 11B** requires fine-tuning to the task [for details, see 15] , whereas **OLMo-2-1124-13B-Instruct** can do zero-shot, in-context learning. Both perform quite well on difficult pronoun interpretation tasks, though T5's accuracy is unsurprisingly higher [6, 16].

Results are shown in Fig. 1. Both models did reasonably well on implicit causality (T5: $r = .770, p < .001$; OLMo: $r = .773, p < .001$) if well below the noise ceiling ($.91$). Interestingly, both models (esp. T5) were more certain than humans (Fig. 1, top), though this might reflect inference parameters. T5 performed at the noise ceiling for implicit consequentiality ($r = .68, p < .001$), but OLMo failed entirely ($r = .02, p = .71$). Follow-up analyses showed that OLMo was not merely failing to notice the differences between the two sentence types.

These findings show that very good performance on difficult pronoun interpretation tasks does not necessarily result in human-like pronoun biases. Of course, the two may be linked humans; it may just be that LLMs process language differently. Alternatively, it may indicate that pronoun biases are an over-and-above adaptation utilized to speed sentence comprehension in humans. Either way, these findings suggest caution in using (current) LLMs as replacements for human subjects.

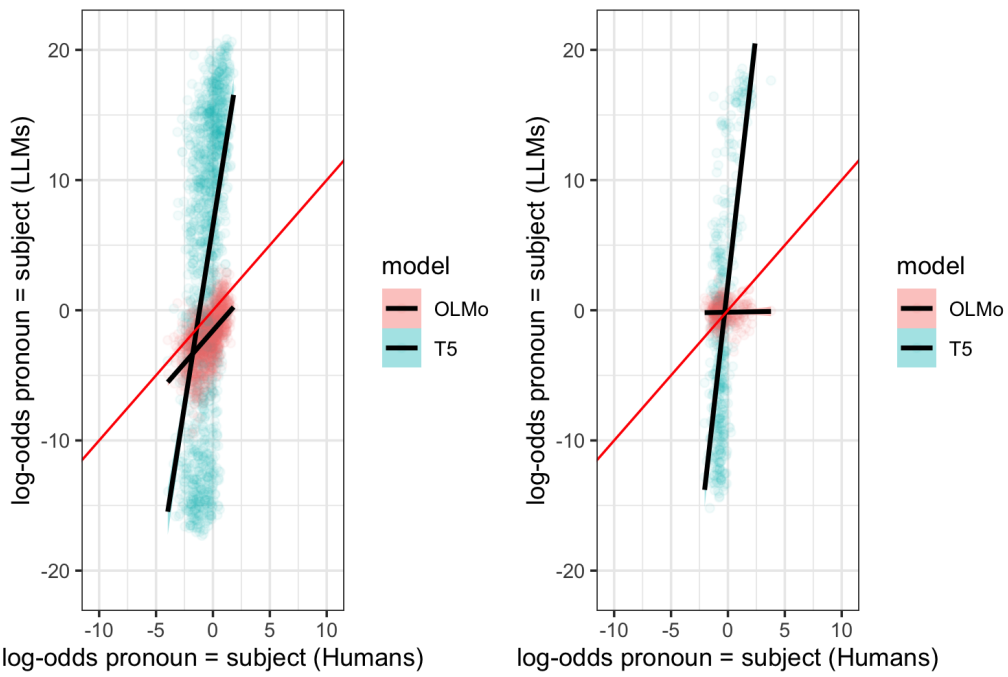

Figure 1: Log-odds of choosing subject for humans (x-axis) and models (y-axis), with implicit causality on the top and implicit consequentiality on the bottom. Positive numbers indicate a subject bias; negative numbers, and object bias. Each stimulus is plotted individually. Separate linear fits are shown for each model, with the 95% confidence interval indicated if not visible. Red line shows depicts what would be a perfect match.

[1] Garvey, C., & Caramazza, A. (1974). Implicit causality in verbs. *Linguistic inquiry*, *5*(3), 459–464.

[2] Hartshorne, J. K., O'Donnell, T. J., & Tenenbaum, J. B. (2015). The causes and consequences explicit in verbs. *Language, cognition and neuroscience*, *30*(6), 716–734.

[3] Stewart, A., Pickering, M., & Sanford, A. (1998). Implicit consequentiality. *Proceedings of The 20th Annual Conference of the Cognitive Science Society.*

[4] Stevenson, R., Knott, A., Oberlander, J., & McDonald, S. (2000). Interpreting pronouns and connectives: Interactions among focusing, thematic roles and coherence relations. *Language and Cognitive Processes*, *15*(3), 225–262.

[5] Kocijan, V., Davis, E., Lukasiewicz, T., Marcus, G., & Morgenstern, L. (2023). The defeat of the winograd schema challenge. *Artificial Intelligence*, *325*, 103971.

[6] Allen Institute for Artificial Intelligence. (2025). Winogrande leaderboard - public submissions [Accessed: 2025-04-30].

[7] Kankowski, F., Solstad, T., Zarriess, S., & Bott, O. (2025). Implicit causality-biases in humans and llms as a tool for benchmarking llm discourse capabilities. *arXiv preprint arXiv:2501.12980*.

[8] Lam, S.-Y., Zeng, Q., Zhang, K., You, C., & Voigt, R. (2023). Large language models are partially primed in pronoun interpretation. *Findings of the Association for Computational Linguistics: ACL 2023*, 9493–9506.

[9] Kementchedjhieva, Y., Anderson, M., & Søgaard, A. (2021). John praised mary because he? implicit causality bias and its interaction with explicit cues in lms. *arXiv preprint arXiv:2106.01060*.

[10] Upadhye, S., Bergen, L., & Kehler, A. (2020). Predicting reference: What do language models learn about discourse models? *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 977–982.

[11] Davis, F., & van Schijndel, M. (2020). Discourse structure interacts with reference but not syntax in neural language models. *arXiv preprint arXiv:2010.04887*.

[12] Ferstl, E. C., Garnham, A., & Manouilidou, C. (2011). Implicit causality bias in english: A corpus of 300 verbs. *Behavior Research Methods*, *43*, 124–135.

[13] Hartshorne, J. K., & Snedeker, J. (2013). Verb argument structure predicts implicit causality: The advantages of finer-grained semantics. *Language and Cognitive Processes*, *28*(10), 1474–1508.

[14] Sakaguchi, K., Bras, R. L., Bhagavatula, C., & Choi, Y. (2021). Winogrande: An adversarial winograd schema challenge at scale. *Communications of the ACM*, *64*(9), 99–106.

[15] Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W., & Liu, P. J. (2020). Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of machine learning research*, *21*(140), 1–67.

[16] Groeneveld, D., Beltagy, I., Walsh, P., Bhagia, A., Kinney, R., Tafjord, O., Jha, A. H., Ivison, H., Magnusson, I., Wang, Y., et al. (2024). Olmo: Accelerating the science of language models. *arXiv preprint arXiv:2402.00838*.